

SIMATS ENGINEERING
ASSESSMENT TOOL II — DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 180 min	Total Marks: 100	Reg No :192324189 Name: N HARSHAVARDHAN

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool: Data Analysis Project

Weightage: 20%

Code of Conduct:

I N HARSHAVARDHAN (192324189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N HARSHAVARDHAN

Criteria	Understanding of Concepts 25%	Analytical Design 25%	Use of Tools 20%	Analysis of Results 20%	Documentation 10%	Total
Q1						
Q2						
Q3						
Q4						
Q5						

DSA0603 — DATA HANDLING AND VISUALIZATION

CO3_AT2_Assessment Tool: DATA ANALYSIS PROJECT

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

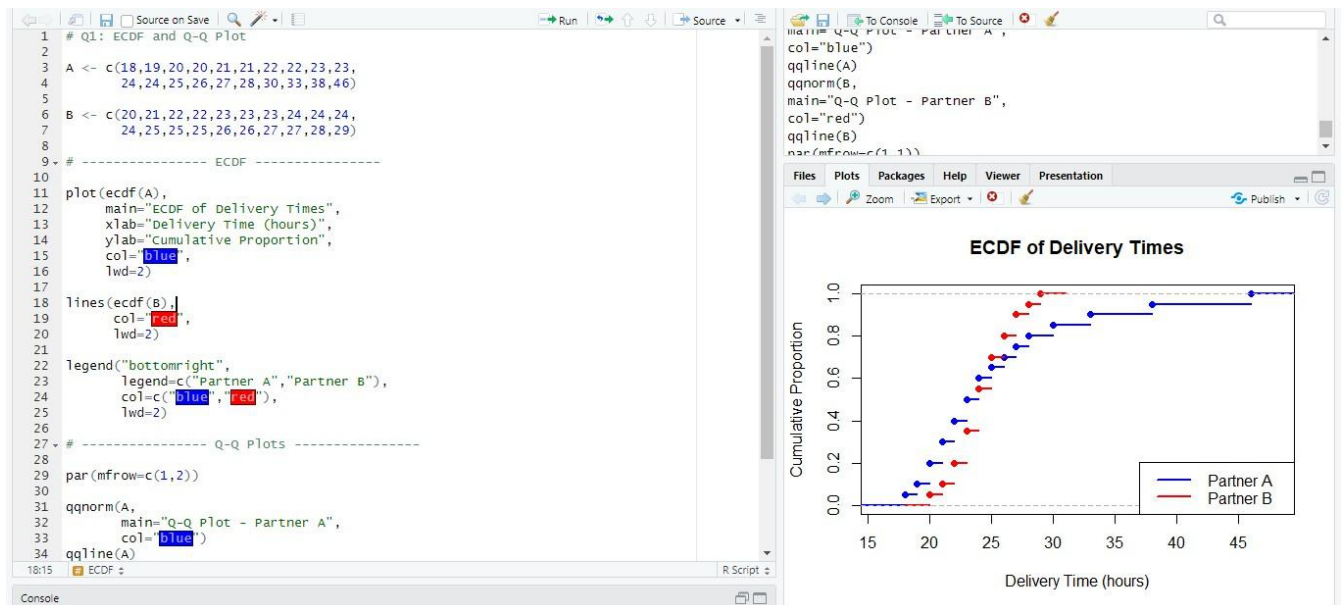
General Instruction: For each project task, use an appropriate visualization in R, justify the design choice, interpret the result in context, and identify potential visualization bias. R/ggplot2 (or an equivalent tool such as treemapify, ggalluvial, or networkD3) may be used. Submit R code, output plots, and a short written interpretation for each question.

Question 1 — Comparing Distributions with ECDF and Q-Q Plots

Scenario: A logistics company records delivery time (hours) for two courier partners over 20 shipments each. Operations wants to know whether one partner is consistently faster and whether either partner's delivery times follow a normal pattern that would justify simple mean-based SLAs.

Dataset:

Partner	Delivery Times (hours), n = 20
Partner A	18, 19, 20, 20, 21, 21, 22, 22, 23, 23, 24, 24, 25, 26, 27, 28, 30, 33, 38, 46
Partner B	20, 21, 22, 22, 23, 23, 23, 24, 24, 24, 24, 25, 25, 25, 26, 26, 27, 27, 28, 29



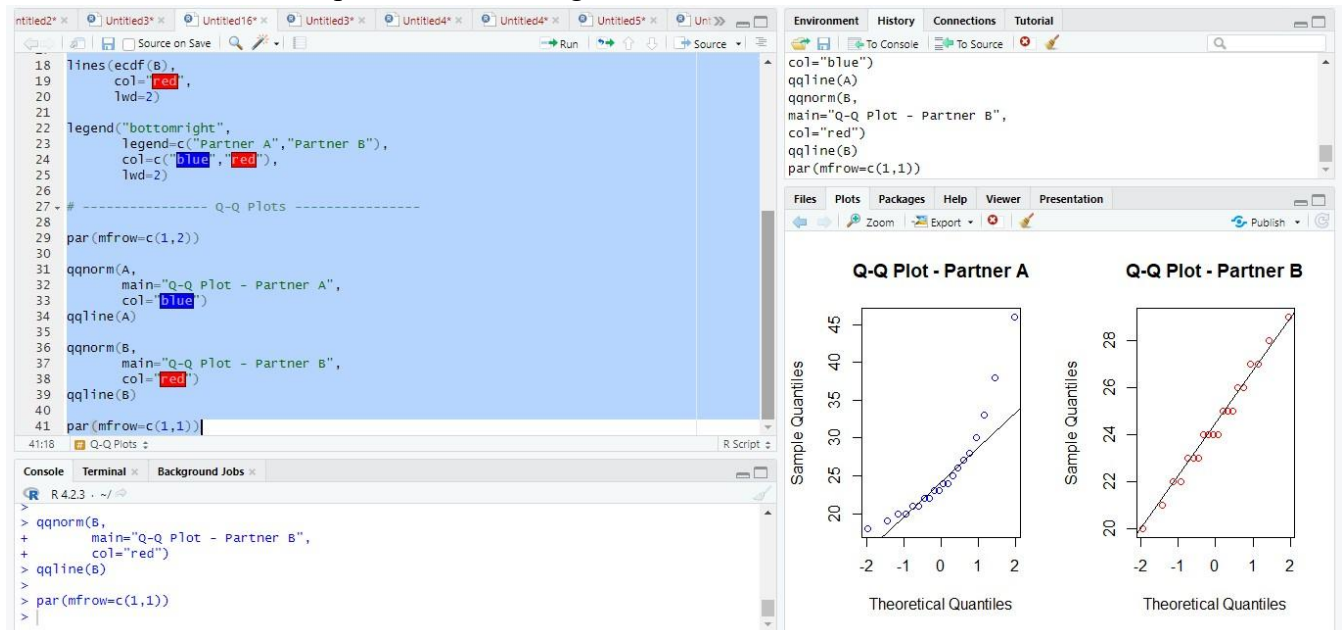
1. ECDF interpretation

The ECDF curves show that **Partner A is generally faster at lower delivery-time thresholds**. The maximum difference occurs around **21–22 hours**, where the ECDFs differ by about **0.20 (20 percentage points)**.

At 21 hours:

- Partner A: 30% completed
- Partner B: 10% completed

So Partner A has a clear operational advantage for faster deliveries.



2. Q-Q plot interpretation

Partner A has strong deviations from the straight line at the upper end because of delivery times such as **38 and 46 hours**. This indicates **right skewness and possible high-end outliers**.

Partner B follows the normal line more closely, although some deviations remain.

3. SLA recommendation

Empirical percentiles are preferable to mean $\pm$ standard deviation.

Reason:

- Partner A has extreme high delivery times.
- The distribution is not perfectly normal.
- Mean and standard deviation can be strongly affected by outliers.
- Percentiles such as P90 or P95 directly represent actual delivery performance.

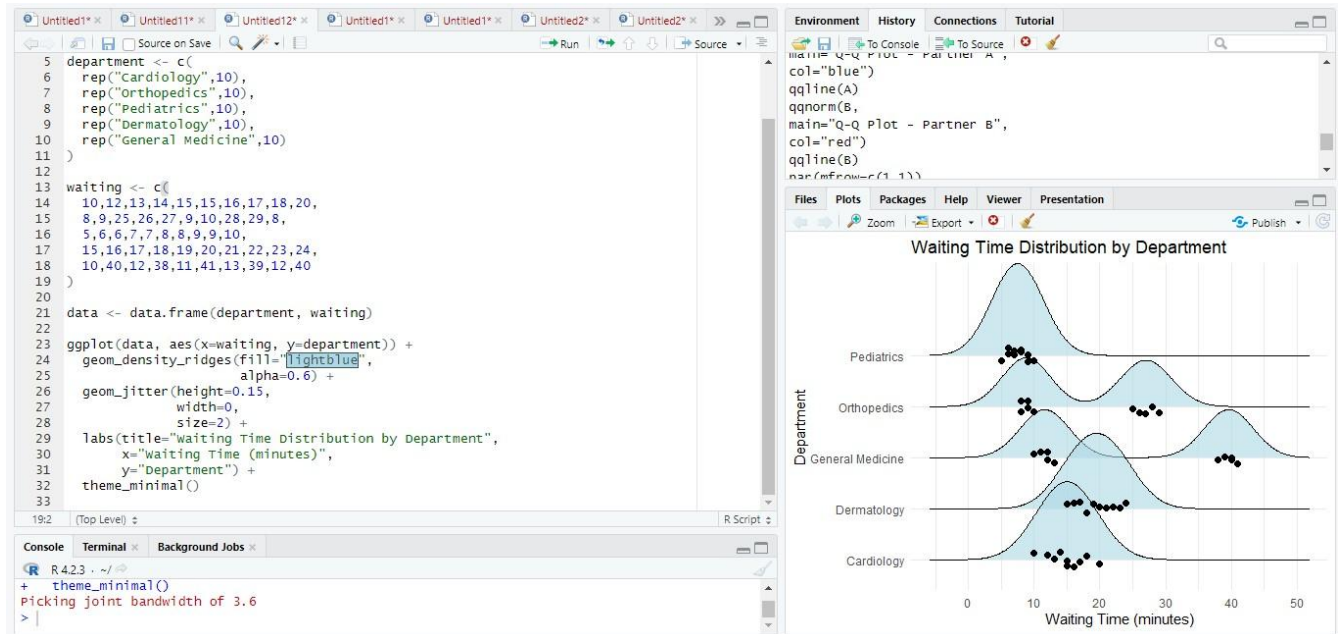
Recommended SLA: Use an empirical percentile such as the **90th or 95th percentile**.

Question 2 — Distribution Shape Across Many Groups using Bean Plots

Scenario: A hospital network measures patient waiting time (minutes) in five outpatient departments over a week. Administration wants to identify which departments have consistent waiting times and which have unpredictable, multi-modal patterns before reallocating staff.

Department	Waiting Times (minutes)
Cardiology	10, 12, 13, 14, 15, 15, 16, 17, 18, 20
Orthopedics	8, 9, 25, 26, 27, 9, 10, 28, 29, 8
Pediatrics	5, 6, 6, 7, 7, 8, 8, 9, 9, 10
Dermatology	15, 16, 17, 18, 19, 20, 21, 22, 23, 24

General Medicine	10, 40, 12, 38, 11, 41, 13, 39, 12, 40
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Interpretation

General Medicine clearly shows a **bimodal distribution**.

Its values are concentrated around:

- 10–13 minutes
- 38–41 minutes

This could indicate two different patient groups, such as:

- routine patients
- complicated/emergency patients

Orthopedics also shows two groups:

- approximately 8–10 minutes
- approximately 25–29 minutes

This suggests that different appointment types or treatment complexity may be responsible.

Variability ranking

From highest to lowest variability:

1. **General Medicine**
2. **Orthopedics**
3. **Dermatology**

4. **Cardiology**
5. **Pediatrics**

Median ranking

From highest to lowest median:

1. **General Medicine – 25.5**
2. **Dermatology – 19.5**
3. **Orthopedics – 17.5**
4. **Cardiology – 15**
5. **Pediatrics – 7.5**

Why are the rankings different?

Variability and median measure different properties.

For example, General Medicine has very high variability because its observations are split between short and very long waiting times.

Bean plot vs boxplot

A boxplot mainly shows:

- median
- quartiles
- spread
- possible outliers

But it can **hide multimodal patterns**.

The bean/density plot reveals the actual shape of the distribution.

Staffing recommendation

A good decision would be:

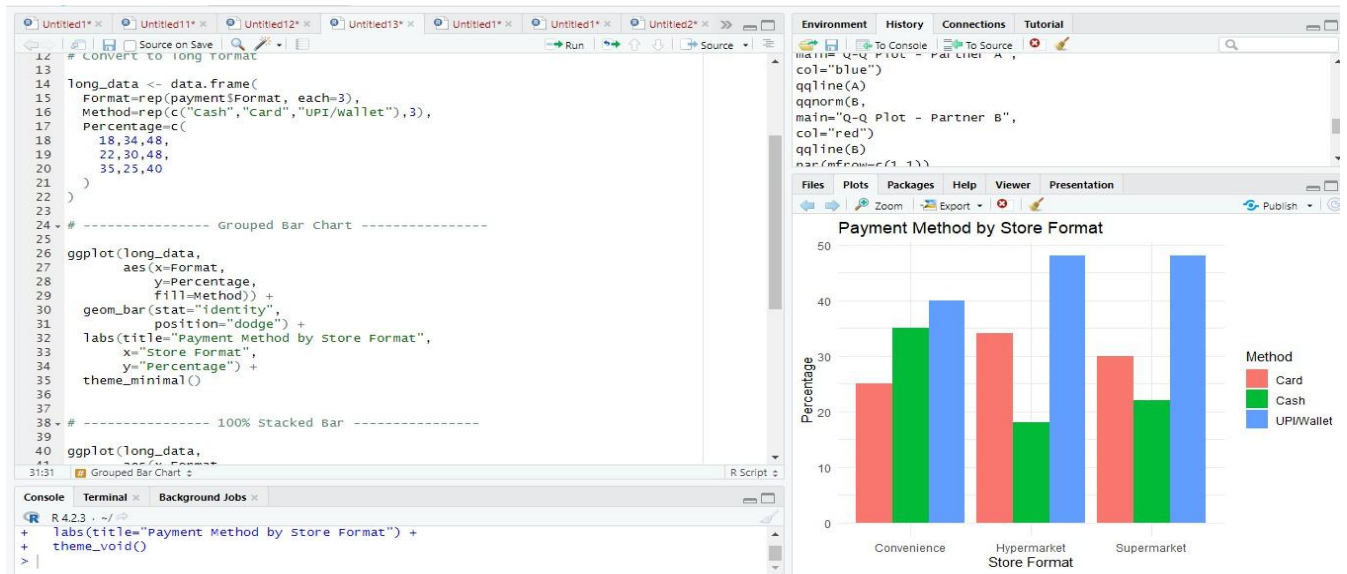
Assign additional staff to General Medicine during periods when complex/high-wait patients are expected, because its bimodal distribution indicates unpredictable waiting times.

Question 3 — Choosing an Honest Proportion Chart

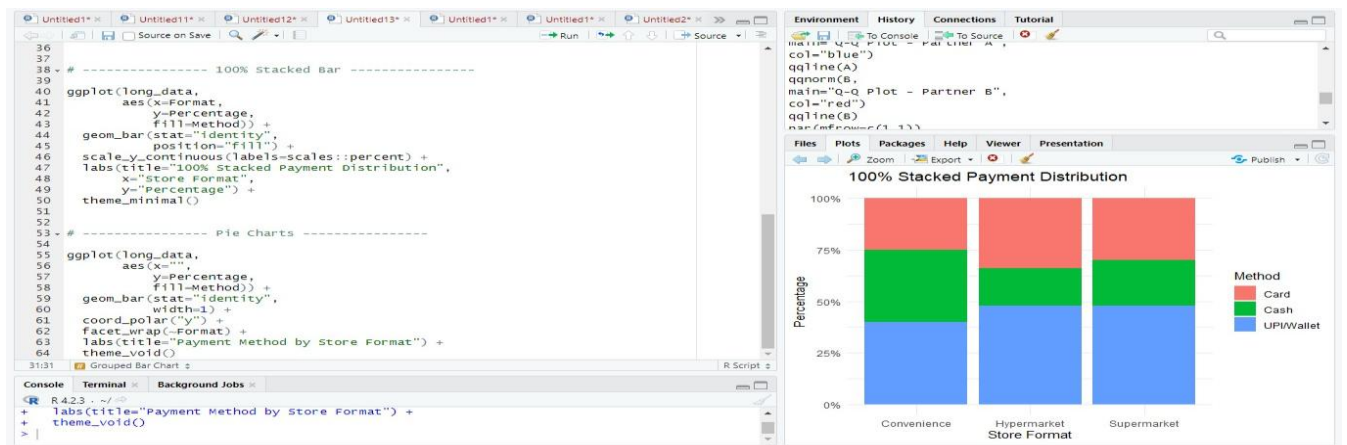
Scenario: A retail chain surveys 300 customers across three store formats about preferred payment method. Senior management wants a dashboard chart that allows accurate comparison of payment-method share across formats without visually exaggerating small gaps.

Store Format	Cash (%)	Card (%)	UPI/Wallet (%)
Hypermarket	18	34	48
Supermarket	22	30	48
Convenience	35	25	40

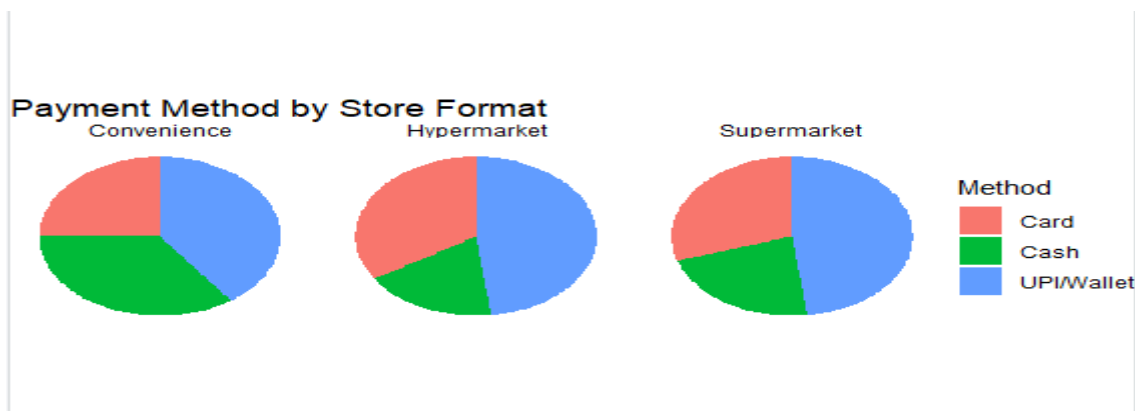
Grouped Bar Chart



Stacked Bar



Pie Charts



Comparison

The **grouped bar chart** is the **easiest** for comparing UPI/Wallet percentages.

UPI/Wallet:

Format UPI/Wallet

Hypermarket 48%

Supermarket 48%

Convenience 40%

Therefore:

Hypermarket = Supermarket > Convenience

The grouped bars make this comparison immediately visible.

Why pie charts are harder

Bar charts use **length**, which humans compare accurately.

Pie charts use **angles and arc lengths**, which are harder to compare precisely.

For example, comparing:

- 48%
- 48%
- 40%

is much easier using bars than trying to compare three circular sectors.

Misleading design

A misleading technique would be to use a **3D pie chart**.

It can make the front slices appear larger than the back slices.

Another misleading technique would be to **truncate the bar-chart y-axis**.

For example, starting the axis at 30% instead of 0% would make Convenience's 35% cash usage appear dramatically larger.

Best design: use a normal 2D bar chart with a zero baseline.

Question 4 — Nested Proportions in a Household Budget

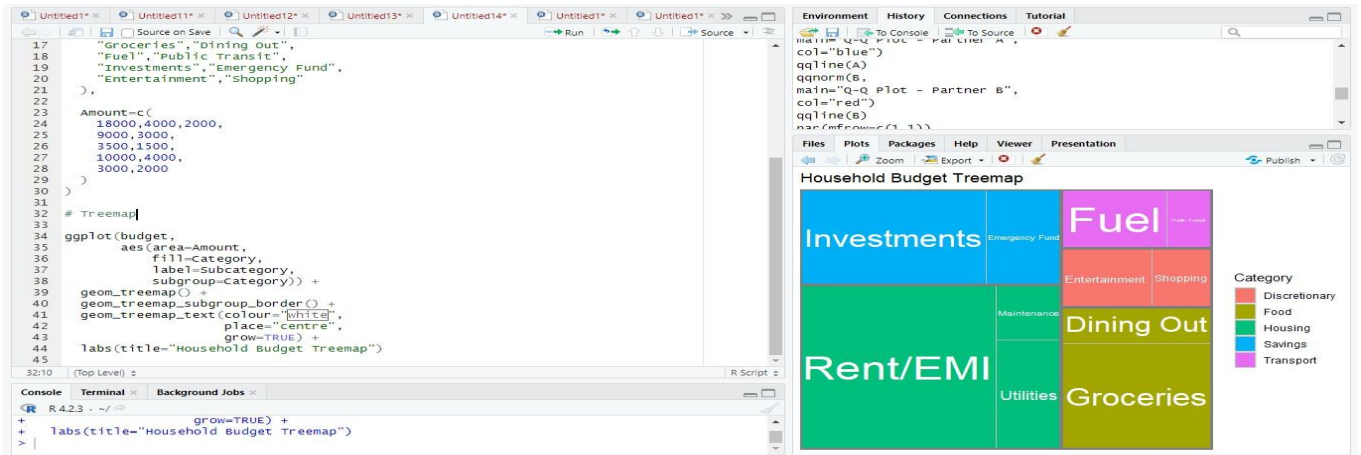
Scenario: A financial-planning app aggregates the average monthly household budget (₹60,000) across spending categories and subcategories for a client segment. The product team wants a dashboard visual that shows both overall category share and subcategory composition within each category.

Category	Subcategory	Amount (₹)
Housing	Rent/EMI	18000
Housing	Utilities	4000

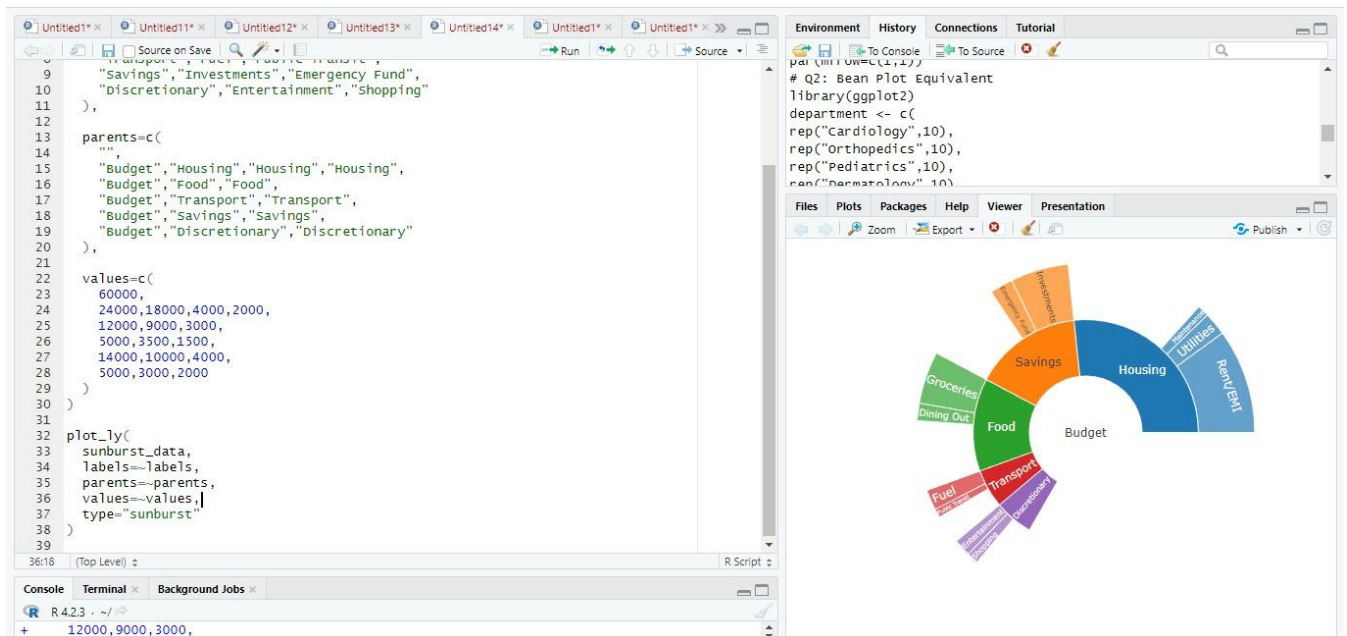
Housing	Maintenance	2000
Food	Groceries	9000
Food	Dining Out	3000
Transport	Fuel	3500
Transport	Public Transit	1500
Savings	Investments	10000
Savings	Emergency Fund	4000

Discretionary	Entertainment	3000
Discretionary	Shopping	2000

Treemap



Sunburst Chart



Housing percentage

Housing:

$$₹18,000 + ₹4,000 + ₹2,000 = ₹24,000$$

Percentage:

$$60000 \div 24000 \times 100 = 40\%$$

Therefore:

Housing = 40% of total budget.

Largest Housing subcategory

Rent/EMI = ₹18,000

$6000018000 \times 100 = 30\%$

Therefore:

Rent/EMI = 30% of the total budget.

Within Housing:

$2400018000 \times 100 = 75\%$

So Rent/EMI represents **75% of Housing spending.**

Treemap vs Sunburst

Treemap:

- Uses rectangular area.
- Good for seeing relative size.
- Efficient use of screen space.

Sunburst:

- Clearly shows hierarchy.
- Category → subcategory relationship is intuitive.
- Comparing subcategories from different categories can be difficult because they appear at different angular positions/rings.

Treemap bias

Area is harder to judge than length.

For example, two similarly sized rectangles can appear different because of their width-to-height ratios. Therefore, a viewer may incorrectly judge two categories as substantially different even when their actual amounts are close.

Question 5 — Flow Analysis using Sankey Diagram and Parallel Sets

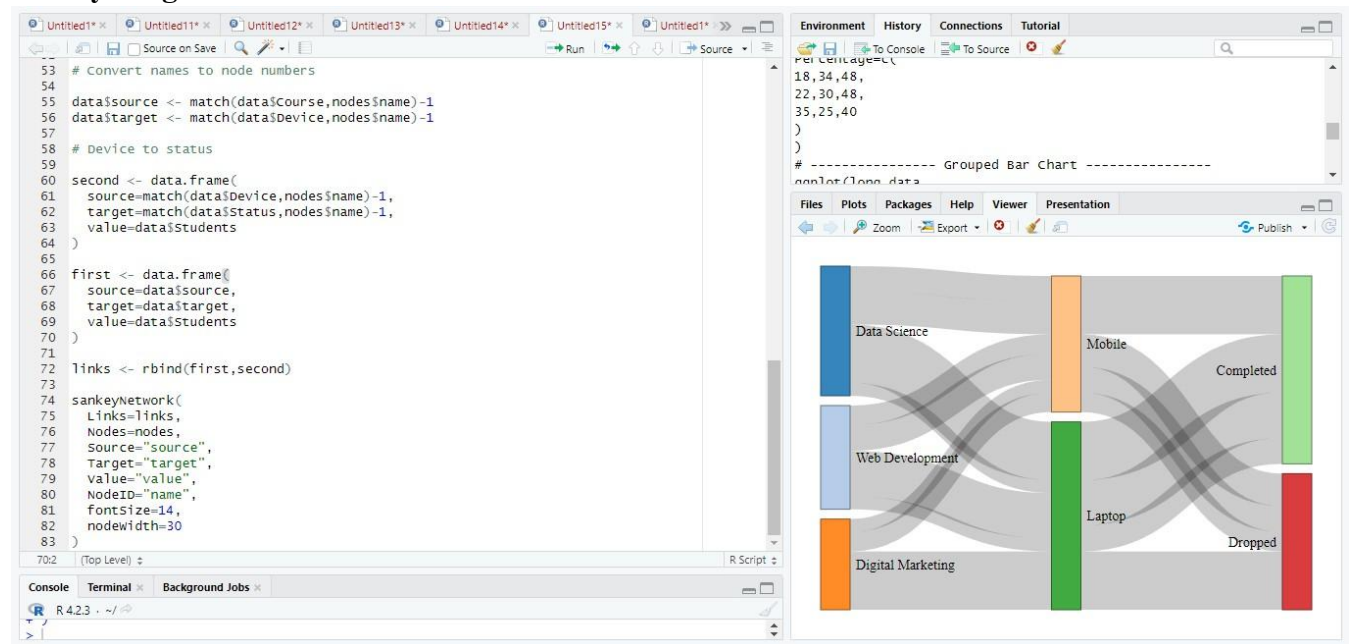
Scenario: An online learning platform tracks 250 students from the course they enrolled in, through the device used to study, to their final completion status. The academic team wants to understand dominant enrollment-to-outcome pathways and spot problem areas.

Course	Device	Completion Status	Students
Data Science	Mobile	Completed	20
Data Science	Mobile	Dropped	25
Data Science	Laptop	Completed	45
Data Science	Laptop	Dropped	10
Web Development	Mobile	Completed	15
Web Development	Mobile	Dropped	20
Web Development	Laptop	Completed	35

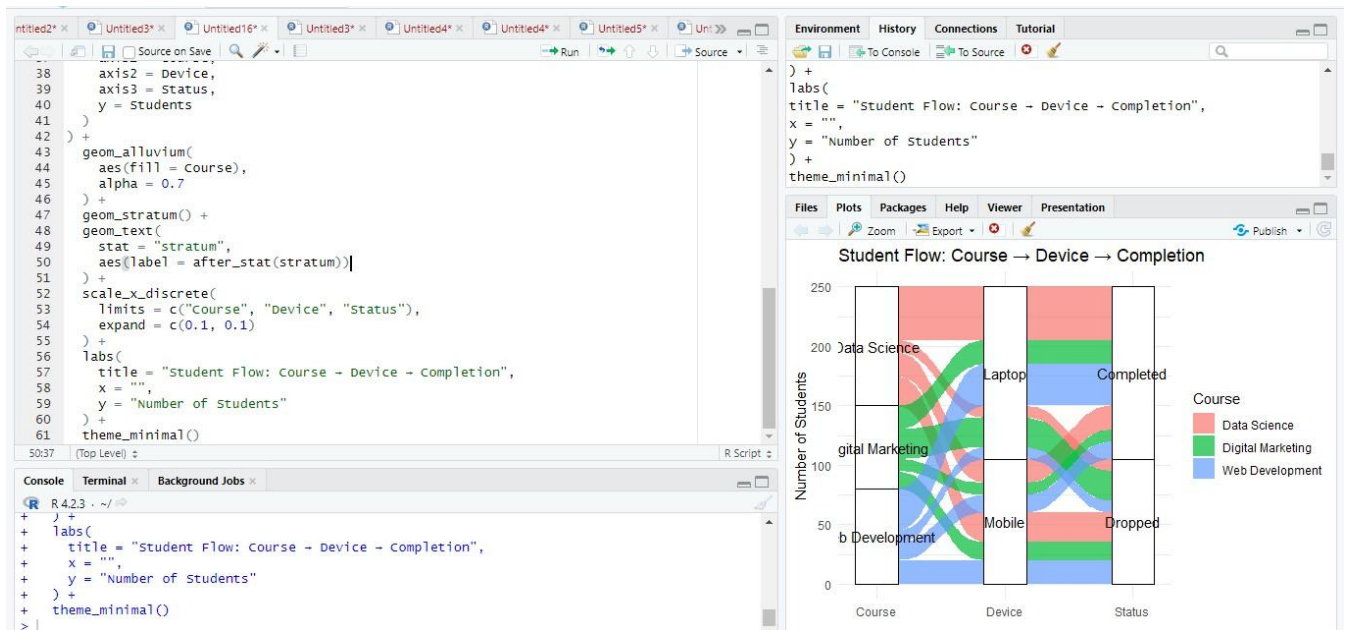
Web Development	Laptop	Dropped	10
Digital Marketing	Mobile	Completed	10
Digital Marketing	Mobile	Dropped	15
Digital Marketing	Laptop	Completed	20

Digital Marketing	Laptop	Dropped	25
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Sankey Diagram



Parallel Sets / Alluvial Plot



Three largest flows

From the dataset:

Flow	Students	Percentage
Data Science → Laptop → Completed	45	18%
Web Development → Laptop → Completed	35	14%
Data Science → Mobile → Dropped	25	10%

There is a **tie at 25 students** for some other flows, including Digital Marketing → Laptop → Dropped. For example:

$$25045 \times 100 = 18\% \quad 25035 \times 100 = 14\% \quad 25025 \times 100 = 10\%$$

Course with weakest completion

Data Science

Total:

$$20 + 25 + 45 + 10 = 100$$

Completed:

$$20 + 45 = 65$$

Completion rate:

$$10065 \times 100 = 65\%$$

Web Development

Total = 80

Completed = 50

$$8050 \times 100 = 62.5\%$$

Digital Marketing

Total = 70

Completed:

$$10 + 20 = 30$$

Completion rate:

$$7030 \times 100 = 42.86\%$$

Therefore:

Digital Marketing has the weakest completion outcome at approximately 42.9%.

The largest problem is visible at the **completion-status stage**, particularly the dropped students.